Chapter 6
Relativity and the Present



The view that now prevails, since Einstein's theory has come to be regarded as an essential part of physics, is that time is an aspect of the universe which depends on the observer.
—Whitrow, *Time in History* (1988, 173).

6.1 Introduction

According to a post in the Associated Press in early 2017, "Scientists have detected a black hole that's taken a record-breaking decade to devour a star—and it's still chewing away. The food fest is happening in a small galaxy 1.8 billion light-years from Earth."[1]—except, of course, that this last sentence is self-contradictory. If the light that astronomers are seeing is from a galaxy 1.8 billion light-years from Earth, then the event described took place 1.8 billion years ago, so to say that the event "is happening" is a stretch, to say the least, as is the statement that it is "still" chewing away. In the same vein, someone wrote to an online site wondering how it could be possible that we can see the Crab Nebula some 6500 light years away, given that Chinese astronomers saw the supernova that gave birth to it less than 1000 years ago—"meaning that light from [that event] shouldn't have reached us yet for us to see it, right?".[2] No, wrong: the light from the supernova would have taken some 6500 years to reach the Chinese astronomers too, so that the supernova would have occurred some 7500 years ago.

What these examples show is just how difficult it is to rid ourselves of the idea of the world existing at an instant. We are aware of the things around us, including the planets and stars as we see them, and for us this constitutes "the universe". But even allowing for the fact that what we see at cosmic distances happened long ago, we still imagine that there is a fact of the matter about what is happening *now*. In classical physics it was assumed that the world-at-an-instant is a well-defined concept. Classi-

[1] "Black hole taking more than a decade to gobble up star", *The Associated Press*, posted Feb 06, 2017 4:22 PM ET.

[2] https://www.quora.com/, accessed August 29, 2017.

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R. T. W. Arthur, *The Reality of Time Flow*, The Frontiers Collection,
https://doi.org/10.1007/978-3-030-15948-1_6

cal spacetime could be thought of as a succession of three-dimensional instantaneous spaces threaded together along the time dimension, constituting a four-dimensional structure, absolute spacetime[3]—at least it could in retrospect, after Minkowski had introduced the notion of spacetime in his reformulation of Einstein's relativity theory. But as we saw in the previous chapter, in the Minkowski spacetime of SR there is no unique world-at-an-instant, because simultaneity is relative to the inertial frame chosen. In the words of Carol Rovelli, "The idea that a well-defined *now* exists throughout the universe is an illusion, an illegitimate extrapolation of our own experience" (Rovelli 2018, 44).

But the idea of the present as something robust, perhaps the very bedrock of our experience of the world, is deeply ingrained in our mental attitudes. At the very least, it is thought, what I am experiencing now is real. This is what Whitehead called "presentational immediacy", "the familiar presentation of the contemporary world" (Whitehead 1930, 13), which he conceived in terms of a moment shared between a percipient and the things perceived. The relativity of the present undermines the idea of such a unique present moment—if that moment is assumed to be without duration, an instant. (This accounts for Whitehead's rejection of such an instantaneous now in which events occur, as discussed in Chap. 3.) It also appears to preclude the doctrine of presentism, according to which what exists is just what exists now, and equally, the common conception of becoming as consisting in the advance of such an instantaneous present from the past to the future, as in the "growing block" model we treated in Chap. 3. But this then raises the question: *what is the status of the present in relativity theory*?

One answer is to deny the present any objective significance, as in the static block model, where the present is relegated to being a mere feature of subjective experience. We have already discussed and rejected this model in Chap. 3, where it was found to commit an equivocation about the reality of events. But closely related to this view is what may be called the *relativized* or *observer-dependent present*, the first of three conceptions of the present offered in response to the relativity of simultaneity that I wish to consider in this chapter:

(i) *the present is relative to the observer's inertial frame.*

On this view, *present existence* is not an absolute notion, but *is relative to the observer's inertial frame*; the world-at-an-instant is a three dimensional, but relative, reality. This is a popular view, especially among physicists, who (taking their cue from Einstein) have concluded that the present is "relative to the observer's coordinate system".[4] Since they have also remained wedded to the idea that becoming

[3] Actually, we need to make a distinction between Newtonian and Galilean spacetime. According to Newton, each point x in absolute space marks the same position through time, making the resulting spacetime a unique structure, absolute space time. But according to the Galilean transformations, a boost in velocity v will make a state of affairs described in terms of the coordinate x indistinguishable from one described in terms of the transformed coordinate $x' = x - vt$. Since v is an arbitrary inertial velocity, this means there is an infinite class of equivalent spaces, each associated with a different value of v, in each of which x' marks the same position through time: this is Galilean spacetime.

[4] Thus Stephen Hawking writes: "In the theory of relativity there is no unique absolute time, but instead each individual has his own personal measure of time that depends on where he is and how

must be conceived in terms of the advance of a world-wide now, proponents of this thesis of the observer-dependence of the present also conceive becoming itself as observer-dependent.

The observer-dependence of simultaneity and becoming was contested by Alfred A. Robb as early as 1911 in a direct reaction to the way Einstein described his theory. Basing his theory on the Principle of Retarded Action, according to which influences must always take a finite time to propagate (see Chap. 4), Robb argued that no distant event could be simultaneous with any given point-event. He therefore insisted that:

(ii) *what is present at a given spacetime point is (strictly speaking) constituted by that point alone.*

This is the *punctual present* view, later taken up by Milič Čapek, and carefully elaborated by Howard Stein in some influential papers. Although formally correct, this conception of the present does not sit well with our experience of objects around us as present to us. Following certain suggestions made by Stein, Steven Savitt and I have independently suggested that a more natural notion of the present in relativity, and one appropriate to an event of finite duration, endows it with a certain spatiotemporal extent. The region of spacetime in question was first defined by the Russian physicist A. D. Alexandrov, so we may call the corresponding notion of the present the *Alexandrov present*:

(iii) *present to an object or process during an interval of its proper time π, are all those processes in the region of spacetime comprised within the absolute future of the beginning of this (usually short) interval and the absolute past of the end of that interval.*

My discussion proceeds as follows. I begin by stressing the need for the distinction between what is in an observer's (visual) experience and what such an observer infers to be 'now'. In fact, even classically, what you see—the events appearing in phenomenal experience—are all events that occurred in the past. This already creates difficulties for the kind of primacy accorded to the events of present experience. So before examining the status of the present in the light of relativity theory, it behooves us to take a step back, and examine pre-relativistic conceptions of the present, the topic of Sect. 6.2. We begin by examining Einstein's own contrast between time in relativity and the classical conception. Contrary to what seems to be implied by his account, the distinction between what *is* happening now and what *is seen as* happening now is not a discovery of relativistic physics, but is already taken into account in classical physics with the Doppler effect.

This occurs only in Einstein's attempt to situate the theory philosophically, and does not affect the substance of his theory. Nevertheless, it is intimately linked with his "interpretation of coordinates and time as products of measurement" through which they are connected with "immediate experience" (Einstein 1954, 247). A similar interpretation in his 1905 paper provoked various misconceived attempts to refute

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he is moving" (Hawking 1988, 33). "Thus time became a more personal concept, relative to the observer who measured it" (143).

his theory on the part of his contemporaries. For according to this epistemological reading, local time is given its meaning, even if mediately, by its relation to the possible experiences of a putative observer. In Sect. 6.3, I show how analogous mistaken conceptions of relativity as "relativity to the observer" underlie the idea of the observer-dependent present. But one of the main lessons of relativity is that there is no instantaneous present "seen" by any observer. As in classical physics, an event *seen* as happening now by an observer in relative motion to it differs from what the same observer would *infer* to be happening now if they were relatively at rest; but in relativistic physics this effect is compounded by the effect of time dilation. I use a formulation of the scenario of the Twin Paradox to show the contrast between what the twins would see and what they would infer classically (because of the Doppler effect) and relativistically (because of the combination of that with the time dilation effect). As we shall see, in relativity distant events simply are not instantaneously *present* to an observer at a spacetime point in any meaningful sense.

A realization of the latter fact underlay the response of Alfred Robb to the problem of the present, which he offered in his (1911) and subsequent publications. In Sect. 6.4 I argue that Robb's view is of particular interest because he shows how Minkowski spacetime can be constructed from an invariant relation of chronological precedence, without the measuring rods and clocks assumed by Einstein. As clarified by later authors, Robb's construction can be seen as built around a principle of local becoming from which, together with the principle of relativity and certain natural constraints on physical process, the Galilean and Lorentz transformations can both be derived (as I sketch in the Appendix to this chapter).

In ascribing fundamental reality to isolated point-events, however, Robb's position is vulnerable to the objections raised in Chaps. 2 and 3 to Grünbaum's attempt to constitute reality from point-events. Robb's construal of the present is further vitiated by his insistence that his instants (point-events) be immediately accessible to consciousness. As I argued earlier, a point-event is better regarded as the bound of a process or event that has some duration. In keeping with this, finally, I argue in Sect. 6.5 that it is more appropriate to characterize the present in terms of what is compresent to such an extended event or process. This gives a view more consistent with our experience of compresence by allowing for an extended time of apperception. It yields an account of the present in the sense that you might say you were present at some significant event; but it is not intended to capture or reconstitute the present of phenomenal experience.

6.2 Pre-relativistic Conceptions of the Present

In a semi-popular essay he wrote in 1936 Einstein gave an account of how time in relativity theory differs from its construal in classical mechanics. He proposed that there are two independent postulates constituting the idea of the objective time presupposed in classical physics. One correlates "the temporal sequence of experiences with the readings of a 'clock', i.e., of a periodically recurring closed system", and

this constitutes the core meaning of what he calls "local time".[5] The second extends this notion to all the events in space, "by which notion alone the idea of local time is extended to the idea of time in physics" (Einstein 1954, 298). The first postulate envisages an observer with a clock, the material embodiment of time; the clock is a measuring device giving local time its "empirical content", a conceptual precedence of measuring instrument to concept that "corresponds exactly to the precedence of the rigid (or quasi-rigid body in the interpretation of the concept of space" (298–99). Prior to the advent of the theory of relativity, Einstein continued, the prevailing illusion was that "the meaning of simultaneity in relation to spatially distant events, and consequently, the meaning of physical time, is a priori clear", and that "this illusion had its origin in the fact that in our everyday experience we neglect the time of propagation of light" (299). As a result, "we are accustomed … to fail to differentiate between 'simultaneously seen' and 'simultaneously happening'", with the consequence that "the difference between time and local time is blurred" (299).

Local time, then, is for Einstein the time of a given observer, established by the linkage between his experiences and the clock he has immediate access to, and then extended across the whole of space by an appropriate measurement procedure. Where classical mechanics erred, according to him, was in supposing that extending any one observer's local time across the whole of space in this way would yield a unique class of simultaneous events, whatever the observer's state of motion—just as to untutored common sense there appears a whole class of simultaneous events that are immediately disclosed to sight. As a result of this blurring of the difference between time and local time, Einstein charges, the "lack of definiteness in the concept of objective time" was "veiled" or hidden" in classical mechanics, where time was represented as "given independently of our sense experiences" (299). Einstein's account implies that objective time should be grounded in the observer's sense experiences by a proper extension of his local time, the clock in his own reference frame, across the whole of space. Once this is done appropriately, it reveals that objective time lacks definiteness, because of its dependence on the observer's own state of motion. Simultaneity is determined by the observer's sending out and receiving reflected light signals, and the results will vary with the observer's own state of motion. Each observer has his own local time. There is no absolute simultaneity.

Here Einstein is drawing a picture of the classical theory of the present by way of contrast with time in relativity theory, and at the same time giving an epistemological account of why the latter theory is superior. The Lorentzian view presupposes that there is one absolute time in which all events occur, and this is plausibly the result

[5] Albert Einstein, "Physics and Reality" [1936]; from *Ideas and Opinions*, (Einstein 1954, 298). Confusingly, this is not the same concept that Lorentz and others called *local time*, which is what results from the extension of local time across all of space, as in Einstein's second postulate here. Of course, we are also familiar with 'local time' in another sense, as remarked by Alfred Robb. This is the sense in which "noon at Greenwich and noon at New York are both described as 12 o'clock local time, although the instants referred to are clearly different" (Robb 1921, 7). But as Robb explains, the idea underlying 'local time' in this sense is that different names are used for the same time at different locations on the Earth's surface, just for the convenience of keeping clocks roughly synchronized with the daylight hours in those locations.

of classical prejudices resulting from our regarding nearby events as simultaneous because they are seen to be so, owing to the very great speed of light. In fact, however, as Einstein would have known very well from his work in the Patent Office, the speed of transmission of light signals must be taken into account in ascertaining the times of distant events. Nevertheless, the way Einstein has set up this contrast with classical physics implies that the classical foundations are faulty in not having taken this into account in determining the simultaneity of distant events, thereby failing "to differentiate between 'simultaneously seen' and 'simultaneously happening'". As we shall see, this is far from the case, and we will need to get this straight to appreciate the classical theory of the present.

Secondly, there is the epistemological grounding that Einstein provides. This 1936 article reflects Einstein's thoughts at a comparatively late period in the development of his thought, some twenty years after he had created his General Theory of Relativity, and over thirty after he had introduced the Special Theory in 1905. Yet the notion of local time he is advocating here, and especially the idea of its receiving its meaning from how it is measured, reveals a continuing commitment to the epistemological considerations that led him to the Special Theory. This is curious given that long before 1936 Einstein had repudiated such Machian considerations as a mere dispensable scaffolding. But it is significant in showing him still wedded to the view that there is no time in relativity theory independent of a subject and his or her state of motion, only a time relative to the observer's coordinate system.[6] This, as we shall see, has resulted in some serious misunderstandings of time in relativity.

Already in 1911, Alfred Robb had objected to the idea that relativity makes simultaneity observer-dependent. Einstein's claim "that events could be simultaneous for one observer but not simultaneous for another moving with respect to the first", was interpreted by him as giving "an air of unreality to the external world which cannot be justified" (Robb 1911, 11). For Robb, Einstein's idea that the simultaneity of spatially distant events depends on the observer's point of view "seemed to destroy all sense of the reality of the external world and to leave the physical universe no better than a dream, or rather, a nightmare." (Robb 1921, v). What Einstein was later to characterize as the "lack of definiteness" of objective time was for Robb rather a sign of the failure of simultaneity to apply to spatially distant events at all.

Robb's "nightmare", it should be noted, is something of a caricature. For Einstein, the "subject" could in fact be any material body in motion. But the importance of Robb's rejection of the subjectivism he saw in Einstein's thinking is that it led him to provide an observer-independent foundation for relativity based on an invariant relation of temporal precedence, and a theory of the present distinct from the classical idea of a plane of simultaneity. Moreover, in the introduction to his 1911 paper Robb appeals to the very distinction that Einstein would later invoke (in 1936) between the time we see an event to occur and the time at which it occurs. So a comparison

[6]Rugh and Zinkernagel make a similar point: "But even if Einstein subscribed less to a positivistic philosophy over the years (Ryckman 2001), he kept referring to the role of measuring rods and clocks in order to give physical meaning to the coordinates in special relativity" (Rugh and Zinkernagel 2009, 7). (The article referred to is updated as (Ryckman 2014).) See Howard (2017) for an analysis of Einstein's philosophy of science.

of Robb's use of this distinction with Einstein's will serve as a convenient way to introduce discussion of the classical theory of the present, and how it should be contrasted with the situation in relativity.

In his paper Robb discusses a hammer blow seen from a distance, and how auditory and visual perceptions of it differ in time "when the occurrence is at a distance from the observer" (Robb 1911, 8). He cites Ole Rømer's experiments from 1675–76 on the eclipses of Jupiter's satellites which showed the speed of light not to be infinite. "Thus the instant of one's visual perception of a distant event cannot be identical with the instant at which the event occurs, and we perceive near and distant events simultaneously which certainly do not occur simultaneously." (8). Robb concludes that, given the finite velocity of light, and the fact that with our other senses we perceive distant events more slowly still, all the distant events we perceive are actually in the past. So, if what is happening now is what is happening at this instant that we are perceiving, then, strictly speaking, "*the present instant does not extend beyond here, and the only really simultaneous events are events which occur at the same place.*" (18).

As described so far, Robb's argument depends only on the finite velocity of light, and not on any features unique to relativity theory. But this means that the same argument could have been conducted classically: nothing that appears to our senses is happening now, save what is coincident with us at that point in space and time. So if what is real is what is now, then reality is restricted to that fragmentary vantage point. Robb himself does not go that far, rejecting the idea that reality does not extend beyond the observer's own immediate perceptions as solipsism.[7] But we do find a non-solipsistic version of such a view of reality advocated over two thousand years ago by the Buddhist sect of Sautrãntikas, mentioned in Chap. 2 in association with time atomism. As explained by the Buddhist scholar Fyodor Shcherbatskoy, they proposed that reality is constituted by fragmentary, instantaneous point-events or *dharmas*, a doctrine known as the "Theory of Instantaneous Being". This is "the leading idea of Buddhism—there is no other ultimate reality than separate, instantaneous bits of existence" (Shcherbatskoy 1932, 1, 80). Although the origins of this view are lost in the mists of time (and may, according to Shcherbatskoy, even predate Buddhism itself), it has taken different forms in the different sects making up the chequered history of Buddhism. Originally it was a metaphysical thesis, devised in opposition to the Realists, who held that space and time were as real as the matter contained in them. But later it was given an epistemological foundation by Dignâga and Dharmakîrti (Shcherbatskoy 1932, 1, 108–114). If the only events that are real are those that are present to our perceptions, then, given that these perceptions themselves take time, a little reflection reveals that every event we perceive as 'now' actually took place earlier, and so is not strictly speaking still in existence. From this they concluded that what is past, future, or imagined is all alike unreal, and only the

[7] "A normal individual who is not a solipsist (and a solipsist could hardly be regarded as a normal individual) believes in the existence of more than his own self and his own perceptions, and one is accustomed to regard these perceptions, under normal circumstances, as representing things as real as one's self but in some sense external." (Robb 1911, 7–8).

instantaneous, punctual present is real.[8] Admittedly, if perceptions are conceived as instantaneous, and light is thought to travel instantaneously, this would not follow: at least everything we see at a given instant could be understood as existing now. But there are indications in the Nyaya texts that the Buddhists, like the Epicureans, held that the transmission of light, although almost inconceivably rapid, is not in fact instantaneous.[9]

One thing this shows is that there is no such thing as a univocal "classical theory of the present" shared by all authors prior to Einstein.[10] But it shows more: it shows that the common tendency to apply the label "real" only to those events that we are perceiving now—the doctrine of presentism—is already deeply problematic in pre-relativistic philosophy. If everything that is real consists only in my present perception of it, as Berkeley assumed with his doctrine of *esse est percipi* ("to be is to be perceived"), then nothing beyond my present perception can be real, including the chairs or trees I see around me, or the self I take to be perceiving them.[11] Even Hume, whose empiricist principles led him to reduce the self to a succession of instantaneous impressions, presupposes a continuous time in which these impressions can form a succession—an idea which cannot be derived from sense impressions.[12] There are similar difficulties for Kant, for whom time is "the form of inner sense". For if there is no continuously enduring self possessing this form of inner sense, contrary to presentism, then there is nothing that could be continuously producing the appearances of things in time.[13]

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[8] Shcherbatskoy summarizes Buddhist presentism as follows: "Everything past is unreal, everything future is unreal, everything imagined, absent, mental… is unreal. Ultimately, real is only the present moment of physical efficiency [i.e., causation]." (Shcherbatskoy 1932, 70–71).

[9] South Asian History, Pages from the history of the Indian sub-continent: Rational and Secular Philosophy, Logic and Epistemology, Philosophical Development from Upanishadic Metaphysics to Scientific Realism: "Nyaya texts on causality indicate that there was an awareness that light travelled at a very high speed but the transmission of light was not instantaneous." (http://india_resource.tripod.com/upanishad.html).

[10] Craig Callender, for example, thinks it obvious that "Traditional conceptions of time all accept the non-uniqueness assumption", which he characterizes as requiring that "at least one event in the universe shares it present with another event's present" (Callender 2000, S592). Although this is perhaps implicit in Aristotle's notion of το νυν, whose identity across space was made explicit by Gassendi and Newton, the example of the Sautrāntikas shows that it cannot be held to be universally true of traditional conceptions.

[11] One might add: since (according to orthodox theology) God does not perceive events "now", Berkeley's stratagem of trying to save the reality of things whose *esse est percipi* by allowing God as a perceiver will not suffice. One is therefore obliged either to deprive God of his atemporality, as in modern process theology, or, like Newton, to allow that God perceives all events in time directly by a kind of proprioception, as if in his own Sensorium; either solution would have been equally heretical to the mind of Berkeley.

[12] This point is made with characteristic wit, irony and modesty by Jorge Luis Borges in his essay (in two versions) "A New Refutation of Time", (Borges 1964, 190–205), in which he combines Berkeley's idealism and Hume's theory of self with Leibniz's Principle of the Identity of Indiscernibles to refute the reality of time.

[13] As we shall see in Chap. 8, Julian Barbour, taking his lead from John S. Bell, argues precisely this: there is nothing continuously producing the appearances of things in time: rather, the appearance

Einstein himself, of course, has often been situated in the empiricist tradition. Logical positivists attributed his success in overthrowing classical conceptions of space and time to his having correctly defined them in relation to experimental procedures. Percy Bridgman, for example, drew inspiration from Einstein for his doctrine of operationalism, according to which the meaning of a physical concept is synonymous with the operations by which we measure it. According to him, the key to Einstein's creation of the Special Theory of Relativity was his insistence on a proper operational definition of the concept of simultaneity (Bridgman 1927). As Robert DiSalle has observed (DiSalle 2006, 110), there is ample support for this in Einstein's original 1905 relativity paper, where Einstein sets out by giving two "operational" criteria for simultaneity: one could define it "by substituting 'the position of the small hand of my watch' for 'time'" (Einstein [1905] 1923, 39). But this will only define time "exclusively for the place where the watch is located" (39), so that for distant events this needs supplementing by a coordinative definition, in terms of "the time at which a light signal from each event reaches the watch" (DiSalle 2006, 110).

Recent studies of Einstein's philosophy of science have downplayed the significance of such positivist elements in his thought, showing how they played little role in his reconceptualizing of space and time, as opposed to its philosophical justification. Notably, however, the two operational criteria Einstein gave in his 1905 paper are the same two criteria for giving meaning to time that Einstein outlines in his 1936 paper described at the beginning of this section. As we have seen, there he insists that "the meaning of simultaneity" of spatially distant events cannot be determined except in relation to how it is measured, so that each observer in a different state of motion would have his own local time. So even though he later repudiates Machism, and insists that the creative formation of concepts is free and largely mathematical, he still insists that the meaning of time is different for differently situated moving observers.

This is not without significance. As I shall argue, Einstein's way of conceiving the meaning of time in terms of observers' possible experiences leads to a conflation of proper time with coordinate time that is still prevalent among contemporary physicists and philosophers. It has influenced not only those who saw themselves as following his lead, but even those who have opposed aspects of his natural philosophy.

A good case in point is provided by Whitehead, who can be regarded as having constructed an experience-based epistemology similar to that described by Einstein, but then used it to drive a metaphysics of time opposed to that implicit in Einstein's theories. Like the mature Einstein, Whitehead was critical of positivistic empiricism. Despite maintaining that philosophy should begin with an analysis of the content of subjectivity, he insisted that it needed to be corrected by reference to what he called the "reformed subjectivist principle", so as to show how objectivity is related to

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of the past is encoded in a "time capsule" at any point in spacetime, yet there is no continuous time threading through these points.

subjectivity.[14] But this conception was substantially moulded by his understanding of relativity theory, where he followed Einstein in holding that it is a consequence of the theory that observers would construct different "meanings" for space and time depending on their state of motion.[15] Like Einstein, however, Whitehead rejected a subjectivist interpretation of this, insisting that these "meanings" are associated with the bodies of the observers, not their minds (Whitehead 1925, 118). As a result there are therefore "multiple time-systems", different "external" spacetime relations specific to observers or objects in different states of motion.

So Whitehead tries to construct time and spacetime from the differing possible experiences of observers. But where Einstein had inferred from his operational definitions that absolute simultaneity is an illusion, Whitehead parted company. For him, simultaneity is a basic component of experience, prior to objective temporal relations of succession: "simultaneity is an ultimate factor in nature, immediate for sense-awareness" (1920, 56). So, faced with the demise of the world-wide instant implied by relativity, his response was "to distinguish simultaneity from instantaneousness" (56), and to deny the instantaneity of simultaneity. "There is no such thing as nature at an instant posited by sense-awareness. What sense-awareness delivers over for knowledge is nature through a period" (57). This, he agrees with Bergson, is *duration*, namely "the whole simultaneous occurrence of nature which is now for sense awareness" (53).[16] It is in such an extended duration that "the creative advance of nature" occurs. A primitive sense of the passage of time is delivered by feelings of causal efficacy, and at a higher level, we conceive time as an extended serial order of moments. The latter is the objective time of physics, an abstract concept, produced by the activity of the experiencing subject, but actualized by the passage of nature.

This is not the place for a detailed analysis of Whitehead's complicated metaphysics of time. As an attempt to save the absoluteness of simultaneity in the face of

[14] Hurley (1986, 99). Although Whitehead interprets the experiencing subject in terms of "actual occasions" rather than the other way around, the world for him, as for Kant, is nothing more than what is disclosed in sense experience.

[15] "But now it seems that the observed effectiveness of objects can only be explained by assuming that objects in a state of motion relatively to each other are utilising, for their endurance, meanings of space and time which are not identical from one object to another." (Whitehead 1925, 120). Ironically, Whitehead sees himself as opposing Einstein's operational definition of simultaneity based on light signals, where "the very meaning of simultaneity is made to depend on them" (119), by one in which "the meanings of our terms" are "physical facts expressible in terms of immediate perceptions" (56). See (Bain 1998) for further discussion. Bain claims that Whitehead's insistence is a *semantic* one: "the manner in which theories are formulated and the types of concepts they employ should make clear what one's ontological commitments are". But the same applies to Einstein: see e.g. Ryckman (2014).

[16] Cf. Bergson: "We claim that a single time and an extension independent of duration continue to exist in Einstein's theory considered in its pure state; they remain what they have always been for common sense" (Bergson 1965, 30); and "While the duration of the phenomenon is *relative* for the physicist, since it is reduced to a certain number of units of time and the units themselves are indifferent, this duration is an *absolute* for my consciousness, for it coincides with a certain degree of impatience which is rigorously determined." (Bergson 1944, 368).

relativity, it simply will not work.[17] But I believe that there is a moral to be drawn from its failure. This is that the strategy of trying to construct a physical theory out of terms whose meanings are given through their being related to an observer's immediate experience is mistaken in principle. It inverts the methodological practice of physics, which does not *construct* its concepts out of subjective or phenomenological givens, but *infers* what is the case independently of knowing subjects. For the present of classical physics is an *inferred* present. Knowing the speed of light and sound and the time it takes to perceive events of various kinds, we can determine which events are those that could possibly be experienced by someone in a given location in classical spacetime, and, knowing the motions of distant bodies we can calculate also those events that are happening at the same time. We can infer, for instance, the position of each of Jupiter's moons at the very time I observe them through my telescope (in slightly different positions) if we know enough about their positions and orbital velocities. We can, that is, infer where they are "now".

This is precisely the consideration on which Rømer's determination of the speed of light is based.[18] He was able to establish the time taken for Jupiter's moon Io to complete its orbit around that planet from a series of observations of its successive emergences from eclipse by Jupiter's shadow. But if light travelled at a finite speed, the times at which it was seen to emerge from eclipse would differ when the Earth was at different distances from Jupiter. So, knowing the time of its emergences on August 7 and August 14, 1676, the times for its emergences in November could be calculated. The observed times differed from those predicted, allowing Rømer to calculate the extra time taken by light to reach the Earth from Jupiter in November 1676 compared to August 1676 as about ten minutes, giving the value of c as 2.2×10^8 m/s, as noted in Chap. 5. Rømer's calculations were confirmed by Fontenelle and later improved upon by Bradley. As was later recognized, it is implicit in these considerations that the frequency of an oscillating body (such as the orbiting Io) will appear to speed up as it approaches you, and to slow down as it moves away. This is the famous Doppler Effect, explained by the Austrian physicist Christian Doppler in 1842.[19] This effect is already noticeable with sound, as the pitch of an approaching emergency vehicle's siren rises at it comes towards you, and then falls as it recedes into the distance.

So the distinction between the perceived present and the inferred present was, as Einstein certainly knew, perfectly well established in classical physics. Aristotle's το νυν, an indivisible now constituting the boundary between past and future, was part of the Western classical heritage (as we saw in Chap. 2). That such a moment extends spatially across the whole world is clearly articulated by Pierre Gassendi in

[17] Whitehead was no armchair philosopher, however: in an effort to give substance to his metaphysics of space and time, he constructed a theory of gravity on that basis that was a serious alternative to Einstein's General Relativity for many decades.

[18] For sources and thorough discussion see (Rømer and Cohen 1940).

[19] Although the idea of the effect is implicit in the calculations of Rømer et al., Doppler gets the credit for first discussing the changes of frequency of waves whose source is in relative motion explicitly in his treatise (Doppler 1842). It was confirmed for sound waves three years later by C. H. D. Buys Ballot.

the mid-seventeenth century: "As any moment of Time is the same in all places, so any portion of Place is in all times."[20] Gassendi's notion of a world-wide moment was subsequently adopted by Newton, for whom "each and every indivisible moment of duration is *everywhere*" (Newton 1999, 941). Moreover, as we saw in Chap. 4, Newton's "absolute true and mathematical time" is distinguished from its sensible measures by the "equation of time"—that is, by the correction due to the relative motions of the bodies involved, the need for which is evidenced empirically "by eclipses of the satellites of Jupiter" (Newton 1999, 410). Newton was under no illusion that light travelled instantaneously, and in fact was among the first to accept Rømer's establishment of the finite speed of light.

Thus it is not the case that classical physics erred in extending the time of any one observer's immediate experience across the whole universe. This universal or absolute time is an inferred time, and its distinction from the times of events as experienced by observers is firmly enshrined in the Doppler effect. This was well known prior to relativity and the discovery of time dilation. The distortion of time involved in time dilation is a distinct effect; but it too is an inferred effect, even though, like the Doppler effect, it will manifest itself empirically.

In sum, we have seen that if we keep in mind the distinction between what is perceived or experienced and what is inferred, there is no confusion in the notion of a world-wide present in classical physics. It is perfectly normal in physics to infer things that go beyond our experience, so long as supposing them helps to explain our experiences, and obtains confirmation through them. The problem with the theory of the present implicit in classical physics is not that it posited the existence of simultaneous events which could not be part of our experience at a given instant. It was, as Einstein himself demonstrated, that the supposition of a unique class of such events turned out to be incompatible with two other well-confirmed empirical results: the independence of the speed of light from the speed of its source, and the principle of relativity.

6.3 The Observer-Dependent Present

Physicists commonly take relativity to mean that all motion is relative to the observer. If you are moving relative to me, then I am moving relative to you with the same relative speed in the opposite direction. But then by the same token, if your length is foreshortened for me, mine is for you, your time is dilated for me, mine for you. On such a view motion is observer-dependent, and so are all the quantities dependent on it. Thus simultaneity is relative to the observer's coordinate system, and the objectively given present of classical physics must be rejected, as Einstein charged. Early critics of Einstein picked up on this implied lack of objectivity. In

[20] "Ut quodlibet Temporis momentum idem est in ominbus locis; ita quaelibet Loci portio omnibus temporibus subest." Pierre Gassendi, *Animadversiones in Decimum Librum Diogenis Laertii* (Lyon 1649, 632); my translation.

1927 W. D. MacMillan, a respected professor and author of a number of physics textbooks, refused to accept the theory. According to him, it "played havoc with the fundamental concepts of the entire human race". He continued,

> Time is no longer a public matter, the same for all, but is a private, personal matter. Your time and my time are different, just as your personality and my personality are different. Two events, one in New York and one in Chicago, are simultaneous in my time, but if you are moving with respect to me these two events will not be simultaneous in your time.[21]

More famously, the astronomer Herbert Dingle, at the end of a distinguished career in which he had served as Astronomer Royal and had written books explaining relativity theory, became an outspoken critic. If each of the twins in the twin paradox scenario is moving relative to the other, he reasoned, then each's life-process will be slowed down relative to the other, because of time dilation. But then each will age less than the other, an obvious impossibility![22] He was unfazed by experimental confirmation of Einstein's theory: "surely, one does not need an experiment to prove that one clock cannot at the same time work both faster and slower than another" (Dingle 1972, 19).[23]

But there is no impossibility here, as we saw in the previous chapter. The difference in the twins' ages when they reconvene is real, not observer-dependent, notwithstanding the symmetry between their situations when they are in relative inertial motion. Indeed, Einstein himself, despite his talk of the "lack of definiteness" of time in relativistic physics, nevertheless regarded time dilation as a physical effect, not just a "personal matter". As we saw, it was correctly predicted by him as a result of special relativity. Indeed, he suggested an experimental test of time dilation, concluding that "a balance-wheel clock that is located at the Earth's equator must run very slightly slower than an absolutely identical clock, subjected to otherwise identical conditions, that is located at one of the Earth's poles" (Einstein [1905] 1923, 904–905). The same goes for Langevin. In fact, it was precisely to try to forestall the kinds of objections

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[21] W. D. MacMillan, "Postulates of Normal Intuition", in (Carmichael 1927, 54). MacMillan (1871–1948) wrote standard textbooks such as *The theory of the potential: Theoretical mechanics* (1958)*, Statics and the dynamics of a particle*, *Dynamics of rigid bodies: Theoretical mechanics* (1960), and *Periodic orbits about an oblate spheroid*.

[22] Dingle (1890–1978) made this objection in his introduction to the translation of Bergson's *Durée et simultaneité* (Dingle 1965, xvi). This was part of a quixotic campaign to refute relativity, culminating in his *Science at the Crossroads* (1972). But although these works appeared a whole generation after MacMillan, Dingle had already made his essential objection to time dilation in *Nature* (Dingle 1939) while still accepting relativity: "In an article in *Nature* I claimed that the twins must necessarily age at the same rate because it was an essential requirement of the special theory of relativity, which I then believed to be sound, that no observation was possible that would enable one to ascribe the motion preferentially to either twin." (Dingle 1972). See Marder (1971) for a thorough debunking of such misunderstandings about relativity.

[23] Similar criticisms were made by Bergson (1965, 74–78) and Lovejoy (1931). In his discussion of them Čapek (1971, 245–256) endorses their wholly mistaken view that "the time-retarding journey is meaningless within the special theory" (246), having applauded Dingle's claim that it is a contradiction for each of two clocks to be time-dilated with respect to the other (241).

made by MacMillan and Dingle that he devised the "Twin Paradox" scenario (and resolved it) in 1911, as we already noted above.[24]

Nevertheless, the idea that the present moment—and thus what is real—is relative to the observer is still prevalent. If the present is observer-dependent, so it is argued, and what is real in an observer's experience is what occurs in the present instant, then what is real can vary with the observer's state of motion.[25] We noted examples of this line of thought in Chap. 5. Thus Hilary Putnam begins his 1967 article with the principle that "There Are No Privileged Observers", defined as follows:

> If it is the case that all and only the things that stand in a certain relation *R* to me-now are real, and you-now are also real, then it is also the case that all and only the things that stand in the relation *R* to you-now are real. (Putnam 1967, 241)

In the context of Special Relativity, he claims, "we have to take *R* to be the relation of simultaneity-in-the-observer's-coordinate-system".[26] Combining this with the principle that "All things that exist now are real", he is able to show (supposing sufficiently many equivalent "observers" at distinct spacetime locations) that all events are equally real. If being real is synonymous with being present to an observer, and there are no privileged observers in the sense just defined, then "the notion of being 'real' turns out to be coextensive with the *tenseless* notion of existence", where the latter "amounts to 'will exist, or has existed, or exists right now'" (247).

Evidently something has gone wrong here. For if any event that exists now is real, and this "amounts to" its existing before now, existing now, or existing later than now, then an event's existing now (say, at a specific time coincident with my utterance) is not distinguished from its existing earlier or later than my utterance; but that is tantamount to denying temporal succession.[27] In depicting reality as a relative to the observer, Putnam has made the same move as MacMillan and Dingle. If all observers are on a par, Dingle argued, then each observer has the same right to say that the other's time is dilated. But since the same clock cannot go faster and slower than another, time dilation cannot be real. An analogous reasoning underpins Putnam's argument: if what is real is what is simultaneous to the observer, then what is real for one observer will be past or future for another. So, since all observers are

[24] The Twin Paradox is often called the "Clock Paradox", and ascribed to Einstein. But as Galina Weinstein has argued, this involves a confusion. What Einstein described and predicted in his 1905 paper is *time dilation*, the retardation of a clock in relative motion. Certainly the very notion of time dilation was regarded as paradoxical by those like MacMillan and Dingle, given the relativity of motion. But it was Langevin who devised the *Twin Paradox* scenario in an effort to forestall such misconceptions. See Weinstein (2012).

[25] Cf. Brian Greene: "*Observers moving relative to each other have different conceptions of what exists at a given moment, and hence they have different conceptions of reality*" (Greene 2004, 133–34).

[26] Here I have silently corrected a mistaken duplication of 'the's in the original: "we have to take *R* to be the the relation…" (Putnam 1967, 242).

[27] This conclusion is somewhat disguised by Putnam's interpreting his position as supporting "the tenseless notion of existence"; for the latter is not supposed to be opposed to distinct events *existing* at different times, to temporal succession, but just to their *coming into existence* at those times. See Chap. 3.

on a par, all these events will also be real. But Putnam should have followed this logic to its conclusion, and inferred that, if the same event can be both in the future (or past) and simultaneous, then temporal succession cannot be real.

The mistake in each case is the same. It is to interpret relativity to an inertial system of reference as being synonymous with relativity to an observer, and then imposing an equivalence of observers' experiences. If two bodies a and b are moving uniformly relative to on another, then all processes occurring on b will be time-dilated relative to a, and all processes occurring on a will be time-dilated relative to b, whether or not there is anyone to observe the dilation. So long as there is relative motion between them, there will be relative time dilation of the processes in one with respect to the other—and this will be so even if we adopt any other inertial frame from which to observe the motions. Although it is true that if we adopt as our inertial frame that in which a given observer is at rest, we will get a picture of how things will appear to that observer, *there is nothing sacrosanct about the observer's rest frame*. Any one observer's frame (any inertial frame) will do just as well to describe the physics, but we should not switch frames while describing the same process.[28]

Just such a switching of frames is appealed to in many accounts of the twin paradox, under the mistaken conception that everything must always be described from a point of view in which the observer is at rest. Such a conception goes along with the idea of a reference frame as defining what is present in the observer's experience. On this reading, what is real in an observer's experience will be all those events simultaneous with the observer at each moment. In actual fact, however, the point-events that are simultaneous with the observer at any instant are precisely *not* events the observer could experience at that time! Here again there is a confusion of the question of what events may be *inferred* to be simultaneous at an instant, with the question of which events *are experienced* at that instant.

We can clarify this by re-examining the twin paradox under the adoption of an Einsteinian cosmos, first on the assumption of an observer-dependent present, and then by carefully distinguishing what the twins *see* from what they *infer*. Accordingly, we now assume that the length of an object moving at speed v relative to an observer is foreshortened in the direction of motion by the factor $\gamma = 1/\sqrt{(1 - v^2/c^2)}$, and times are dilated by the factor $1/\gamma = \sqrt{(1 - v^2/c^2)}$. The mathematics is pretty straightforward. We suppose one twin is Terence, who stays on Earth for twenty years, while his astronaut sister Astrid travels to Alpha Centauri (taken as 6 light years away) at a speed of $0.6c$, and back again to Earth at the same speed. Ignoring any periods of acceleration or deceleration, according to Terence, his twin Astrid is away for exactly twenty years, ten years out, ten years back. Things are otherwise

[28] Cf. Howard Stein: "one often characterizes such notions as simultaneity, time interval, and length, in Einstein's theory, as 'relative to the observer'. As we shall see, this way of speaking can lead to serious confusion. One should not lose sight of the fact that any observer, regardless of his state of motion, can express the results of his own measurements (and those of colleagues) in *absolute, physically meaningful language*, which is *the same for all observers*." (Stein 1968, 12); cf. also "an observer's state of motion does not impose upon him, according to relativity theory, a special view of the world's structure. The view that it does—that the observer has some distinguished subjective or perceptual relation to "his present"—leads only to confusion." (Stein 1970, 293–4).

for Astrid. At such great speed, the distance between the Earth and Alpha Centauri is for her foreshortened by a factor $\sqrt{(1 - 0.36)} = \sqrt{(0.64)} = 0.8$. So she needs to make a journey outwards of only 4.8 light years. An easy calculation shows that, travelling at a speed of $0.6c$, she does it in $4.8/0.6 = 8$ years according to her clock. The distance home is equally foreshortened, so that she takes only 16 years for her journey (still imagining zero turn-around time). Thus when they are reunited and compare their clocks, they find that Terence has aged 4 years more than his sister!

You can see why Dingle found this paradoxical. If all inertial motion is relative, how can there be an absolute difference in their lifetimes resulting from it? While the twins are in relative inertial motion, each's duration will be running slower from the perspective of the other's rest frame. In each leg of the journey, Astrid would infer processes to be happening more slowly on Earth as it receded from her or approached her at $0.6c$: her eight years would correspond to an inferred duration of processes on earth of only 6.4 years!

There is, of course, an asymmetry in the scenario. In order to reunite with her brother, Astrid must turn around, and this requires that she should accelerate. So something must happen at the point at which she switches direction that explains the difference in their lifetimes. For so long as the twins are in inertial motion relative to one another, each twin must indeed infer that the other's clock is running slow. Thus it is often said that the reason for the discrepancy in the twins' ages is that whereas Terence is in inertial motion throughout, Astrid is the one who really moves because of her acceleration, although this acceleration lies outside the scope of the theory. This has led many to claim that a proper resolution of the paradox must appeal to General Relativity, since Special Relativity (SR) applies only to systems in inertial motion.[29] There are indeed statements by Einstein to this effect, and these are probably the ultimate source of numerous similar such claims that can readily be found.[30] But we know this to be incorrect. Special Relativity is perfectly applicable to accelerated motions, which are represented as curved worldlines in Minkowski spacetime. Granted, in the toy model we are considering, the path of Astrid is not so much curved, as kinked. An instantaneous turnaround is unphysical, but this does not alter the fact that we can model it in Minkowski spacetime; also, one can make the same case with a curve instead of a kink.

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[29]For example, Max Born held this to be the case in his (1922, 216). Similarly, J. S. Bell writes that "accelerated observers are not considered in the 'special' theory" (Bell 1987, 77), and Milič Čapek that "It is this acceleration which places the '*voyage au boulet*' clearly beyond the purview of the special theory. Thus only within the framework of the general theory can Langevin's spectacular space trip be correctly analyzed." (Čapek 1971, 246). Čapek's conflation of proper time with time in an observer's rest frame vitiates his attempt to make sense of aspects of Bergson's failed attempt to come to terms with the special theory.

[30]Thus Brian Greene (in the course of an otherwise excellent exposition) writes that "in special relativity Einstein's main focus was on a special kind of motion: constant-velocity motion. It was not until 1915, some ten years later, that he fully came to grips with more general, accelerated motion, through his general theory of relativity." (Greene 2004, 51). One still finds similar statements on the Internet, e.g.: "However, this resulted in a limitation inherent in Special Relativity that it could only apply when reference frames were inertial in nature, (meaning when no acceleration was present)." (http://en.wikipedia.org/wiki/Inertia; accessed January 30, 2018).

But let us return to the idea of the observer-dependent present. If we assume that what is present (and therefore real) for Astrid is what is simultaneous with her in "her own frame of reference", then there must be some dramatic shift occurring at the point of her switching direction. What is real for her at the instant before she turns around at Alpha Centauri—all the events simultaneous with her at that instant—is radically different from what is real for her the instant she begins the return journey. We can calculate how it changes. The instant before she turns around, at the very end of her 8 years of travel, Terence's clock will read $8 \times 0.8 = 6.4$ years. Similarly, during her journey home Terence's clock will advance only 6.4 years, yet it will read 20 when she returns to Earth. So, at the instant she sets out on her journey home, his clock must read $20-6.4 = 13.6$ years. Assuming an instantaneous turn-around, events on Earth that are real for her would have had to have jumped 7.2 (=13.6–6.4) years from one instant to the next.

So on this account, the explanation for the discrepancy in the twin's ages is due to the fact that the acceleration discontinuously skews Astrid's temporal orientation. If we were to follow Putnam's informal way of speaking, we would say that Astrid "experiences" 7.2 years on Earth going by in an instant: events that were "present according to her co-ordinate system" are discontinuously displaced 7.2 years into the past of "her-now" according to that same system.[31] In actual fact, however, no such wrenching change of her experience of any events would occur. She does not *see* any of the distant events that are simultaneous with her, either before or after her turn-around. There is nothing "present according to her co-ordinate system". These are facets of a sloppy use of the ideas of "observer's reference frame" and the observer's "present", and a failure to distinguish between the time an observer might *infer* an event to occur from when the observer would *see* it occurring.[32]

In fact, it will be worth going over the whole thing in some detail to see how this could be the case, but this time carefully distinguishing what an observer sees from what that observer infers.

First, let's examine the case of the interstellar twins from the point of view of classical physics. We assume, as did Rømer, that light travels (from the standpoint of Earth) at a finite speed, and that everything takes place in Newton's absolute time. Astrid will take 10 years to arrive at Alpha Centauri as before. But because light takes 6 years to travel from Alpha Centauri to Earth, when Terence actually sees the event of Astrid's arrival there, 16 years will have passed since they parted! He sees Astrid's clock register only 10 years while his has registered 16. Thus Astrid's clock is will appear to be running slow by a factor 5/8 compared to his, as he will have been noticing throughout her trip, with his telescope trained on her clock. This is the

[31] Cf. also Rietdijk's talk of two observers mutually at rest "experiencing the same 'present'" as one another. As Howard Stein has remarked, "there is of course no such 'experience': the fact that there is no experience of the presentness of remote events was one of Einstein's basic starting points" (Stein 1968, 16, n. 15).

[32] Cf. Sklar (1974, 272): "One must always be careful in special relativity to distinguish what an observer actually sees, literally, from what he computes to be the case"—the sloppiness of this slippage from one to the other perhaps constituting what one of my students called "the fallacy of the slippery slop"!

classical Doppler effect, as noted above. Now, since his sister takes the same 10 years to make her return trip, she is back after 20 years, and thus after only 4 more years by his clock. But when they reunite, their clocks will agree that 20 years have past. So during this 4 years he watches his sister's clock running at $10/4 = 2.5$ times the speed of his own. Thus as Terence views Astrid's return trip and all the processes happening in it, he sees them appearing to occur four times as fast (2.5 divided by 5/8) as during the trip outwards!

Astrid has an analogous experience. When she arrives at Alpha Centauri, she observes Terence's clock to be reading only 4 years. For the image of Terence's clock registering 4 years travels the 6 light-years to Alpha Centauri to arrive there 10 years later (since light is travelling at speed c in Terence's frame of reference). So Astrid sees Terence's clock has been running 2.5 times as slowly as hers (i.e. at 2/5 speed)! But in the ten years it takes her to return home, Terence ages $20-4 = 16$ years, and accordingly on the way home she sees his clock to be running 1.6 times as fast! Thus she, too, is puzzled to see her twin's clock going 4 times as fast on her journey home (1.6 divided by 2/5) as it was on the outgoing leg of the journey.

All this is in accord with the classical Doppler effect. Events and processes occurring in a frame of reference in motion towards an observer appear to be speeded up ("blue shift"); occurring in a frame of reference in motion away from an observer, they appear to be slowed down ("red shift"). None of this has anything to do with time dilation. Everything here is purely classical, and the slowing down and speeding up of clocks observed by the each twin is an appearance only, completely explained by the Doppler Effect. Each twin, moreover, is able to infer what is happening "now" at any given time, and to distinguish it from what he or she is observing as happening now at that time. For instance, Astrid knows that at the moment she reaches Alpha Centauri, Terence's clock will be recording 10 years as having passed, even though she sees it recording only 4. The distinction between what is happening now and what is seen as happening now may be "blurred" in our unreflective everyday experience, but it was not blurred in classical physics, as one might take Einstein's discussion to have implied (Einstein 1954, 299).

Now, not everything about the twin's experiences is symmetrical. Although each twin sees the other's clock running 4 times as fast on the return trip as it was on the way to Alpha Centauri, they did not observe the same degree of red shift while moving apart, and the same degree of blue shift when moving back towards one another—Terence saw a 5/8 red shift while moving apart and a 5/2 blue shift while they were approaching, whereas Astrid saw a 2/5 red shift and an 8/5 blue shift. That they didn't observe the same shifts is explained by the fact that we have assumed that the light is travelling with the speed c in Terence's frame, not Astrid's. But if all inertial motion is relative, they should have experienced exactly the same degree of red shift while moving apart, and the same degree of blue shift when moving back towards one another. So this gives a nice illustration of the predicament facing physicists at the beginning of the twentieth century. All the experimental evidence seemed to suggest that the speed of light is independent of the speed of the source. But this is incompatible with the requirement that all inertial frames are on a par with respect to the laws of physics, unless something else gives. As we saw in Chap. 5, various physicists realized that

appearances would be preserved if the length of a body were foreshortened in the direction of its motion through the aether by a factor of $\sqrt{(1 - v^2/c^2)}$. We have already seen how that resolves the situation about what they infer. But what do they see?

When Astrid arrives at Alpha Centauri, Terence's clock registers 16 years—10 years for her to make the trip, and 6 years for the image of her arrival to reach his telescope on Earth. But, because of time dilation, when Astrid arrives her clock actually registers only 8 years, so the image of the clock hand at 8 years is what Terence sees. So her clock *appears* to Terence to be running slow by a factor of 2, i.e. to be going half as fast as his; and it would have been appearing to go at half speed throughout this leg of the journey. On the return leg, Terence sees his sister's clock advance 8 years whilst his only advances the remaining 4 before their reunion; so the clock (and the aging processes of his sister and everything moving with her) *appear* to be running fast by a factor of 2. Despite these appearances, of course, Terence can *infer* that in each case the effect is 0.8 times what would be expected from the Doppler effect alone: a lag by a factor of 5/8, multiplied by 0.8, gives 1/2; a speeding up by a factor of 2.5 times 0.8, yields 2. Thus he *infers* that Astrid's clock is running slow because of time dilation.

Astrid, on the other hand, on looking back to Earth as she is arriving at Alpha Centauri 8 years later, sees the Tellurian clock register 4 years, as before. By her reckoning, the image has travelled eight years to get to Alpha Centauri, so Terence's clock *appears* to Astrid to be running slow by a factor of 2. On the return leg, Astrid sees her brother's clock advance its remaining 16 years whilst hers only advances 8; so the clock (and the aging processes of her brother and everything moving with him) appear to be running fast by a factor of 2. Again, she can calculate that since the effects should have been 5/8 and 2.5 if they were due to the Doppler effect alone, the difference is due to the fact that Terence's time is slowed relative to hers by the time dilation factor 0.8. (During her 8 years on the outward leg, she sees Terence's clock move 4 years when it should have moved 5 by the Doppler effect alone, since 8 times the Doppler effect of $5/8 = 5$; on the way back she sees Terence's clock move 16 years instead of the 20 that would be 8 times the Doppler effect of 2.5).

Thus the situation is entirely symmetrical: while they are in relative motion, each twin suffers (relative to the other) an *inferred* time dilation, a slowing-down of the aging process. Moreover, each twin suffers precisely the same time dilation relative to the other while they are moving inertially relative to one another. The time dilation is a *real* but *relative* effect. And in terms of what *appears*, both see their twin sibling's clock running slow by a factor of 2 while they are moving apart, and running fast by a factor of 2 when they are approaching one another.

This shows us that the scenario depicted is entirely consistent. But how does it resolve the paradox? If everything is symmetrical, then why don't the twins age by the same amount? The correct explanation, as we saw in Chap. 5, is that it is not any difference between inertial frame-times that accounts for the difference in the twins' ages, but the difference in their paths through spacetime, the measure of which is proper time. It is the proper time elapsing along a particular path in spacetime that measures how fast the processes traversing that path are going, how fast the people or things undergoing them are aging, how fast they are becoming—and *it is this*

proper time that is the time told by the clocks throughout the twins' journeys. In the non-Euclidean metric of Minkowski spacetime, it is the longest, not the shortest, interval between two spacetime points that is given by the straight line in spacetime connecting them—contrary to what a diagram of their motions might suggest. Astrid travels along two sides of a spacetime triangle, and Terence by the remaining side. In each case Astrid's path appears longer when we draw it, but the time elapsed (the integral of the proper time along the path) is shorter. It follows that it can't be said, as one often reads, that the duration of processes in SR is relative to an inertial frame. In the sense of time lapse that is relevant to the twin paradox—how much time elapses for each twin—*it is simply false that time lapse is frame-dependent*, i.e. that it depends on the inertial frame adopted. Indeed, the duration of each twin's journey through spacetime is an invariant measure: the proper time is numerically the same in all inertial reference frames, provided we use the same frame of reference to describe the whole journey.

Thus time does not lapse differently for the twins because of any difference they see in what events are present at any given instant—precisely because they cannot see events that are simultaneous with them at an instant! Whitehead, as we saw, followed Einstein in claiming that different observers would ascribe different "meanings" to time on the basis of their possible experiences. But as David Mermin notes, "That no inherent meaning can be assigned to the simultaneity of distant events is the single most important lesson to be learned from relativity." (Mermin 2005, xii). Similarly, Howard Stein: "the fact that there is no experience of the presentness of remote events was one of Einstein's basic starting points." (Stein 1968, 16, n.15). Distant events simply are not *present* to an observer at a spacetime point in any meaningful sense.

6.4 Robb and the Punctual Present

This brings us to the third of the construals of the present in relativity theory outlined above: what is present at a given spacetime point is (strictly speaking) constituted by that point alone. As we have seen, this view was first articulated by Alfred A. Robb in 1911, six years after Einstein's original 1905 paper, and only three years after Minkowski's.[33] It has since been endorsed by Čapek (1966), (1975), and also by Stein (1968) (although here without attribution to Robb), in their critiques of the arguments of Putnam and Rietdijk discussed in Chap. 5.[34] As we have seen, Robb took exception to Einstein's proposal that "events could be simultaneous for one

[33] Robb (1914); his (1936) is essentially a second edition of this book. Robb had previously published a draft of part of his theory in the short tract *Optical Geometry of Motion* (1911), and later gave a simpler exposition of his (1914) without proofs of the theorems in his (1921).

[34] "Like Rietdijk," objects Čapek, "Putnam retains the old notion of the universal present spread as a 'world-wide instant' across the whole universe, and uses this notion in order to conclude that, in a sense, *everything* is present" (1975, 612–13). But this neglects "the one essential idea of relativity that … 'Here-Now' can never be extrapolated to 'Everywhere-Now'" (613). Similarly, Stein objected that "if 'presentness to each other' of events is taken to mean that *each has for the*

observer but not simultaneous for another moving with respect to the first" (Robb 1936, 11; cf. 1921, v). Consequently, he regarded instants as restricted to spacetime points, and "avoided any attempt to identify instants of time at different places."[35] His instants are point-instants, and his events are point-events.

Robb presents his work as deriving directly from that of Larmor and Lorentz rather than Einstein.[36] This is a little eccentric. He cedes priority to Einstein and Minkowski for the mathematics, but objects to the subjectivism he sees as implicit in Einstein's way of presenting relativity in terms of observers, rods and clocks. For in his own construction of relativity theory Robb makes no use of rods or clocks at all.[37] Neither did Minkowski, of course. But whereas Minkowski's formulation of relativistic spacetime was premised on Lorentz invariance under Einstein's interpretation,[38] Robb derives the whole theory—spacetime and Lorentz transformations—from first principles. Moreover, Minkowski took Einstein's proposal that the local times corresponding to different inertial systems are ontologically equivalent to mean that time, like space, is a mere projection onto spacetime. Time is accordingly stripped of its former pre-eminence, as Minkowski proudly announced: "space by itself and time by itself recede completely, to become mere shadows". In Robb's system, by contrast, time takes pride of place: "the theory of space becomes absorbed in the theory of time, spacial relations being regarded as the manifestation of the fact that the elements of time form a system in conical order: a conception which may be analyzed in terms of the relations of after and before." (Robb 1914, 9)

Let me begin with a brief outline of Robb's theory and its significance for the reality of becoming in SR, before proceeding to his theory of the present.

In his 1914 theory, his "chronogeometry",[39] Robb takes as his starting point relations that for Minkowski are features of the "absolute world", consequences of the structure of spacetime. These are what Robb calls the "absolute relations" of earlier and later, described by Minkowski in the words: "Each worldpoint within the past lightcone of O is necessarily always earlier than O, each worldpoint within the future lightcone is necessarily always later than O."[40] Now what is in the future lightcone represents all the worldpoints that can be reached from O, and what is in the

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other already become, then … in Einstein-Minkowski space-time *an event's present is constituted by itself alone*." (Stein 1968, 15).

[35]This is Robb's description of his 1911 tract in (Robb 1914, 3). In that work, as we shall see, Robb even wrote of one's being "directly conscious" of instants (1914, 8).

[36]"Although generally associated with the names of Einstein and Minkowski, the really essential physical considerations underlying the theories are due to Larmor and Lorentz." (Robb 1914, 1)

[37]"It is quite unnecessary, in our theoretical investigations, to introduce clocks or measuring rods at all." (Robb, unattributed; quoted from Winnie 1977, 134).

[38]As Minkowski writes, "The credit of first recognizing clearly that the time of one electron is just as good as that of the other, that is to say, that t and t' are to be treated identically, belongs to A. Einstein." (Minkowski [1908] 1952, 82; cf. 2012, 117).

[39]The term 'chronogeometry' is apparently due to A. D. Fokker, who happened on substantially the same insights as Robb without apparently knowing of his work. See (Fokker 1953/55), and (Alexandrov 1967, 1119).

[40]Minkowski ([1908] 2012, 119). Minkowski's calling them "necessarily always" earlier and later is doubly unfortunate: as we saw in Chap. 3 the relations could be said to hold "always" only if there

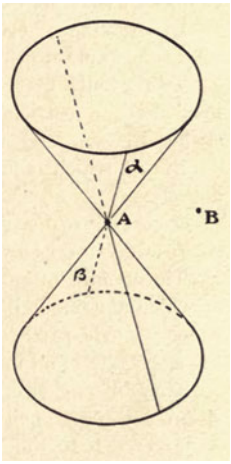


Fig. 6.1 B in the 'elsewhere' of A (from Robb (1914, 5)

past lightcone all those worldpoints from which any process could reach *O*—since no physical influence can propagate faster than light. Robb now takes the possibility of a physical influence going from *A* to *B* as a necessary and sufficient condition for *A*'s being before *B*. Thus one instant *A* is *absolutely before* or *chronologically precedes* another, *B,* if a physical influence can be propagated from *A* to *B*. In such a case, *B* is *absolutely after A*:

> Thus if I can send out any influence or material particle from a particle P at the instant A so as to reach a distant particle Q at the instant B, then this is sufficient to show that B is after and therefore distinct from A. (Robb 1921, 11)

We may call this the Principle of Chronological Precedence, or CP. As can be seen, it presupposes the Principle of Retarded Action discussed in Chap. 4, according to which every physical process takes a finite quantity of time to be completed. Note that so long as CP holds for the propagation of any physical influence, it will not matter whether light or anything else actually travels with the limiting velocity.[41]

As Robb showed in 1914, this means that—restricting temporal relations to these absolute relations only—a given event can be related in order of succession to any event in its future or past light cones, but cannot be so related to any event outside these cones (in what came to be called the event's "Elsewhere"). There are therefore pairs of events that are not ordered with respect to (absolute) before and after, such as the events happening at the instants **A** and **B** on Robb's "Fig. 6.1". The event **B**, being too far away from **A** for any influence to travel between them, is neither before nor after **A**.

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is another time apart from spacetime; and since one event may chronologically precede another contingently, qualifying it as 'necessarily' earlier is inappropriate.

[41] Robb acknowledges that "it may or may not be strictly true of light" that it travels with this limiting velocity, but he takes light to serve for this purpose in his construction.

For example, **B** could be the event on some planet in the Andromeda Galaxy that Paul Davies asked us to imagine, in the Elsewhere of me at the instant **A** when I am considering it. It is true that by walking this way and that I could describe that event as being in the past or in the future according to the time coordinate associated with the frame of reference in which I am at rest. But that event is not present to me in the sense of being a possible part of my experience. It bears no absolute temporal relation to my considering it. And thus, according to Robb, we cannot say that the two events stand in a relation of succession. Neither is earlier than the other. All the events I experience, on the other hand, will be either before or after one another, and therefore distinct. In fact, they will occur in a linear order. They will lie on what Minkowski called my worldline.

There is nothing unique about my worldline, however. On pain of solipsism, what goes for me goes for any other possible observer (this is the counterpart in his theory to Putnam's "No Privileged Observers").[42] Thus if we regard time as constituted by these absolute relations, time as a whole does not have a linear order: not all events can be ordered on a line proceeding from past to future, even though two events that are in each other's elsewhere (i.e. lying outside each other's cones) will be in the past of some event that is suitably far in the future of both of them. In this way, all events can be temporally ordered, even if not every pair of events is such that one is in the past or future of the other. This is Robb's "conical order". In the language of the theory of relations, it is a strict partial order, rather than a serial order.

One could regard all this as little more than a restatement of what Minkowski had already presented as a consequence of special relativistic spacetime. But Robb's insight was to see that by beginning with this relation of chronological precedence CP so defined, together with postulates defining its properties, the whole mathematical structure of Minkowski's spacetime could be derived without assuming anything else. As we saw in Chap. 4, Leibniz had shown how to construct a theory of temporal order on such a basis, a "causal theory of time" (in which he was followed by several others). But Robb's construction also delivered the affine structure of Minkowski spacetime (basically the system of timelike, spacelike and null lines), as well as grounding congruence between intervals and a metric, all on this foundation. Crucially, he achieved this without making any reference to clocks or measuring rods.[43]

Although Robb's own construction was somewhat daunting,[44] other theorists have picked up on the essence of his approach to give equivalent results using more

[42] Thus Robb: "A normal individual who is not a solipsist (and a solipsist could hardly be regarded as a normal individual) believes in the existence of more than his own self and his own perceptions, and one is accustomed to regard these perceptions, under normal circumstances, as representing things as real as one's self but in some sense external" (Robb 1936, 7–8).

[43] Objecting to Einstein's postulating of the existence of clocks to define times Robb wrote "It does not appear a satisfactory mode of procedure to found a philosophical theory upon a complicated mechanism like a *clock* without any precise definition of what constitutes equal intervals of time." (Robb 1914, 8).

[44] "I have been told more than once," Robb ruefully admits in the Preface to *The Absolute Relations of Time and Space*, "that my book [his 1914 *Theory of Time and Space*] is difficult reading." (Robb 1921, vi). There are 21 postulates, the last of which (equivalent to Dedekind's postulate of continuity) finally appears on p. 330! John Winnie gives a thorough examination of Robb's construction in his

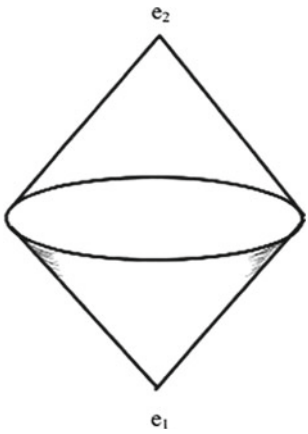


Fig. 6.2 The Alexandrov interval between e_1 and e_2

economical constructions. In a paper of 1967 the Russian mathematician Alexandrov showed how the topology of Minkowski spacetime is uniquely determined "by the propagation of light or, in the language of geometry, by the system of the light cones", noting the equivalence of this derivation to Robb's derivation on the assumption of chronological precedence. The idea is to take as the basic spacetime interval between two point-instants all those point-instants that are absolutely after the earlier instant e_1 and absolutely before the second e_2. This gives the double-cone structure that is now known as the Alexandrov interval (Fig. 6.2).[45]

Other aspects of Robb's theory can be derived with great economy by taking a group-theoretical approach. Thus in his (1964) E. C. Zeeman took such an approach to prove that "Causality Implies the Lorentz Group": that is, that the transformations of spacetime that respect the principle of retarded action are the Lorentz transformations together with rotations in space. Moreover, although Zeeman, like Robb, assumes that the limiting velocity of physical processes is the velocity of light, c, it is possible to construct relativity theory without assuming Einstein's Light Postulate. You need only suppose that there is a finite limiting velocity k, greater than zero (which is infinite in the classical limit). This was first proved algebraically by Vladimir Ignatowski in his (1910), since when essentially the same result has subse-

(1977), relating it to those of Alexandrov and Zeeman mentioned below. As he remarks, "Given Robb's ambitious aims, and the conceptual economy of his foundations, the complexity of his constructions was inevitable. … Only after a few hundred theorems do coordinates appear, and with them the Minkowski metric in its standard form." (Winnie 1977, 158).

[45]The *Alexandrov interval* is discussed by Winnie in his (1977, 156–57), with a diagram depicting this interval for two chronologically connectible events. As Winnie explains, Alexandrov intervals have a profound foundational significance, since a topology for Minkowski spacetime may be defined taking these intervals as basis. This is discussed in detail in (Hawking and Ellis 1973, 196ff.), who show that they are also a sufficient basis for defining topologies of spacetimes in general relativity, provided the *strong causality* condition is met. See also (Savitt 2009).

quently been rediscovered several times. In particular, it can be derived very neatly using group theory.[46]

I sketch such a derivation in the Appendix to this chapter. One supposes versions of the principles of inertia and relativity, together with certain natural assumptions about the symmetries appropriate to displacements produced by boosting a frame: namely, linearity, symmetry under space-reflection, an equal times principle, and "causality", here embodied in the requirement that there exist (at least some) time intervals which do not change sign under any boost transformation. This is the constitutive assumption underlying Robb's principle of chronological precedence, CP. It is then proved that there are only two possibilities consistent with these assumptions. Either the "rapidity"—Robb's term for the additive parameter φ characterizing these boosts—reduces to the classical (instantaneous) velocity, $\varphi = v$, and we recover the Galilean transformations; or $\varphi = k \tanh^{-1}(v/k)$, k is determined empirically to equal c, and we obtain the Lorentz transformations.

This derivation of the Lorentz transformations throws light on the status of *tachyons*, putative faster-than-light processes. It is often said that the existence of tachyons is not ruled out by the Lorentz transformations: just as c is a limiting velocity which slower-than-light processes could never achieve, so it is a lower limit for faster-than-light (superluminal) processes. Nevertheless, tachyons are typically dismissed on philosophical grounds: tachyonic processes, it is objected, give rise to causal paradoxes. For a superluminal process going from one spacetime point p to another q extends across a spacelike interval of spacetime. It follows that if it goes from p to q in one inertial frame, so that p causes q and is therefore earlier than q according to the time-coordinate t of that frame ($p <_t q$), then one can always find a different inertial frame with time-coordinate t' in which q causes p (q is earlier than p) in that frame, ($q <_{t'} p$). This, it is said, violates "causality", the principle that the cause must always precede the effect. But this way of phrasing the objection proceeds as though causality were some optional add-on to the pre-existing structure of Minkowski spacetime. It is not. The violation of "causality" is more profound than this, and it is worth trying to understand why.

First, we may note the objection of Lévy-Leblond and Provost that transformations from one inertial frame to another are parametrized by rapidity, not velocity. Rapidity, however, ranges between $-\infty$ and $+\infty$, and is not defined for velocities faster than c. Thus "there is no rapidity, i.e., no change of reference frame, associated with superluminal velocities" (Lévy-Leblond and Provost 1979, 1045). So the assumption that there can be a transformation from one superluminal inertial frame to another in Minkowski spacetime is without foundation: such transformations would not be part of the Lorentz group characterizing Minkowski spacetime. But we can say more. In our derivation of the Lorentz transformations "causality" is captured by the requirement that there exist (at least some) pairs of events a and b with a before b,

[46] See for example (Lévy-Leblond 1976), (Lévy-Leblond and Provost 1979). Group theoretic approaches are not fashionable among philosophers of physics, since it is known that they are inadequate to characterize the more general spacetimes occurring in the general relativity theory. This explains the preference for intrinsic characterization of spacetimes in terms of manifolds and structures defined on them.

such that no transformation can reverse their order—in Robb's terms, such that a is *absolutely before* b. As discussed in Chap. 4, this is not the assumption of causality itself (causation could in principle be instantaneous); rather it is mandated by the principle of retarded action. The point is that a process is something that proceeds from a state or event a to a state or event b: it cannot do this without taking time, nor can it do it by going from b to a. In this sense "causality" is a principle of becoming: it determines what counts as a process. And in Minkowski spacetime, only pairs of events in timelike or lightlike separation satisfy this principle. In short, a continuous series of point-events from p to q in spacelike separation, such as a tachyon is supposed to be, is simply not a process. To regard it as one is to contradict one of the assumptions constitutive of Minkowski spacetime.[47]

Now let us turn to Robb's construal of the present in relativity. For him an instant is a primitive concept; it is an element of time of which I may be directly conscious.[48] The instants of which I am conscious form a linear series, but instants as a whole need not. They can form a strict partial order, where if neither of two instants is before the other, they are not therefore identical. Robb illustrates this possibility using cones, as we saw above; but at this point talk of cones is only illustrative, prior to the derivation of Minkowski spacetime. He then proposes that a necessary and sufficient condition for an instant B to be after an instant A is that it be "be abstractly possible for a person, at the instant A, to produce an effect at the instant B" (Robb 1914, 7). For the producing of an effect Robb is thinking of "an influence or material particle" being sent from A to B. The instants, as noted, are point-instants, and the events happening at them are strictly instantaneous events such as "two particles striking one another" (6). Now if an instant C is a finite distance from A then such an influence must take some finite time (by the Principle of Retarded Action), so that C is necessarily distinct from A. But simultaneous events are those happening at the same instant (6). It therefore follows that "the only events which are really simultaneous are events which occur at the same place" (6).

The punctual present, however, is very problematic. To say that what is compresent with an event (such as a person considering) is merely what shares the very same spacetime point is hardly compatible with our normal intuitions of presentness. It almost collapses Robb's position into a temporal atomism like that of the Sautrãntikas, as mentioned above. But unlike them, Robb did not insist that only observed instants could exist, and in order to ensure that the instants a person could be conscious of would form a continuous series, in his 21st postulate he explicitly posited a Dedekindian continuum of them. But his position is still susceptible to a version of one of Zeno's paradoxes, as we saw in Chap. 2 in association with Russell's views and Grünbaum's ontology of point-events. As was argued there, temporal becoming can no more take place in an instant than can motion. Therefore, if becoming takes

[47] Kent Peacock has shown (2018, Chap. 4) that it is possible to derive "Lorentz-like transformations for the superluminal case" (54). But the superluminal "motions" so defined will not only not form a Lorentz group, they will not satisfy CP, so I cannot agree with him that they qualify as possible processes.

[48] "An element of time is called an instant and is to be regarded as a fundamental concept. Of any two elements of time of which I am directly conscious one is after the other." (Robb 1914, 4).

place in the present and all that is present at some point is what is at that spacetime point, then, since there is no becoming in an instant, there is no temporal becoming. As we saw, Russell proposed something like this argument as an objection to becoming. Yet his solution to Zeno's parallel argument against motion is that, although there is indeed no motion in an instant, this does not refute the reality of motion, since this consists in a body's having a different position at a later instant. By parity of reasoning, I argued, although a Zenonian argument shows that a process cannot be composed of point-events, it does not preclude there being a process whereby something becomes different at a later time from what it was at an earlier one.

This last consideration, in fact, points the way to an acceptable construal of becoming, and one that "saves the phenomena" of our experience of the present too. This is achieved by recognizing that becoming occurs over a short (even arbitrarily short) duration; even the event of a person considering or apperceiving another event cannot be strictly instantaneous.

6.5 Compresence and Local Becoming

This leads us naturally into consideration of what might be considered present with respect to such an extended event or process. From a phenomenological perspective, this is hardly a new idea. As has long been recognized, events given in our experience, such as our experience of a snatch of melody, are heard as already laid out in order—in this case, as a succession of notes. This is at odds with the above discussion of relativity, where the events are assumed to be instantaneous, and the instantaneous present assumes the possibility of being consciously aware of them at an instant. Obviously, these are the typical abstractions of a mathematical physicist, as is usually acknowledged by qualifying phrases such as "properly speaking" and "strictly speaking". But these abstractions are also responsible for the gulf between mathematical theory and our palpable experiences of events.

Several authors have proposed that this discrepancy between theory and experience can be bridged by introducing the concept of the *specious present*.[49] We already encountered such a view in our discussion of Bergson and Whitehead in Chap. 3. The idea, as originally developed by William James, is that the present or now as we cognize it in practice is "no knife-edge, but a saddle-back with a certain breadth of its own."[50] By this means, our intuitions of presentness as comprising brief processes

[49] See in particular the discussions of Dobbs and Broad (1951), and (Dobbs 1951). Dobbs builds on the speculations of Eddington in (Eddington 1946) about the two-dimensionality of time, as well as Russell's discussion of the paradoxes associated with the specious present in (Russell 1948).

[50] James writes: "In short, the practically cognized present is no knife-edge, but a saddle-back, with a certain breadth of its own on which we sit perched, and from which we look in two directions into time. The unit of composition of our perception of time is a *duration*, with a bow and a stern, as it were—a rearward- and a forward-looking end. It is only as parts of this *duration-block* that the relation of *succession* of one end to the other is perceived." (James 1890, 609–610; quoted from Čapek 1971, 133). Broad (1938, 281ff.) finds others' attempts to define the specious present

and also as encompassing a considerable spatial extent can be preserved. For we do not have to restrict our notion of contemporaneity to what is present to a *point-event* or *instant*, but can apply it to a small extended event of apperception.

Such a construal of contemporaneity has been discussed by Howard Stein in a couple of very influential papers (1968, 1991). The first of these was offered as a rebuttal of the views of Putnam and Rietdijk that we considered above. In a footnote in the first of these Stein provides a definition of contemporaneity that he considers appropriate for Minkowski spacetime in terms of *mutual communication or influence*:

> for processes of more than instantaneous duration, a meaningful and intuitively satisfying notion of 'contemporaneity' can be defined: two such processes may be said to be contemporaneous if part of each is past to part of the other—in other words, if mutual influence ("communication") is possible between them. (Stein 1968, 15 n. 14)

Amplifying on this construal of contemporaneity in terms of interaction in his later article, Stein suggests, as a "plausible anthropological hypothesis", that our "intuitive" notion of the present is grounded in mutual communication, and "first arises 'naturally' in the course of human development and socialization" (Stein 1991, 159). He writes:

> Let us consider a "specious present" π of some percipient being; and let us call an event e "contemporaneous" with π if signals—interaction—influence—can occur *mutually* between e and π. In the Newtonian case, the spatial extent of the set of events contemporaneous with a given specious present is infinite; and it is rather natural to see in this fact the precise correlate, in the physical theory, of the "intuitive" notion of a "present" throughout all of space. (Stein 1991, 159)

Here Stein is linking the notion of mutual communication with what kinds of interaction are countenanced by the physics in question. In the case of Newton's theory, gravitational interaction is assumed to be instantaneous. Kant had already exploited this connection between simultaneity and instantaneous action at a distance in his Third Analogy of Experience. The infinite spatial extent of the hyperplane of simultaneity is grounded in physical interaction. In order for events to be experienced as simultaneous, they must be conceived as being in mutual interaction, thus accounting for the grounding of the instantaneous interactions of Newton's theory.

Perception, however, does not occur by gravitation. So if we are trying to account for how human beings have a conception of the present as including remote events, gravitational action is not to the point. Our intuitions about the present are garnered from what we sense as compresent, and this is dominated by our sense of sight, as noted above in connection with Rømer, Einstein and Robb. Now given the finiteness of the speed of light, and also its value, Stein's suggestion permits us to redeem the notion of the present as having a spatial extent that is, in fact, extremely large. As he explains, it follows that

> the "graining" of time with respect to which a percipient organism can experience conscious interaction with its environment must be such that the "moments" of time (the specious

unintelligible, and gives his own. See Davies (1995, 265–278) and Dainton (2010, 103–120) for more contemporary discussions.

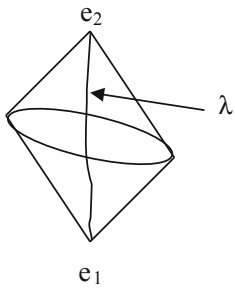


Fig. 6.3 The Alexandrov present, ALEX (e_1, e_2, λ)

> presents) are long enough to allow such signals—and, therefore, light signals—to travel very many times the maximum spatial dimensions of the organism together with its (relevant) environment. (Stein 1991, 161–2)

This explains at one stroke not only why the specious present is of roughly the length it is, but also why humans would naively suppose that the spatial extent of the present—i.e. what we regard as compresent to us during such a specious present—is indefinitely large.

Implicit in this account of Stein's is a notion of the present that can be extended to the relativistic case. For if *a* is the extended event of our becoming aware of some other extended event *e* during some short interval π of proper time, then both *a* and *e* will be short processes, and accordingly both will be represented as segments of world lines in a Minkowski diagram. On this basis, a definition of the extended present appropriate for Minkowski spacetime was independently proposed by myself and Steven Savitt as a natural generalization of Stein's argument. Mine ran:

> the *present* of a process during an interval of its proper time π (for instance, that during which conscious experience is laid down) is that region of spacetime comprised within the absolute future of the beginning of this (usually short) interval and the absolute past of the end of that interval. (Arthur 2006, 148)

In terms of Minkowski spacetime, the absolute future is the forward lightcone, and the absolute past the backward lightcone. The result is that the present of an object during an interval of its proper time is a region of spacetime finite in extent, the intersection of the two "cones", as in Fig. 6.3. Because the region is that which Alexandrov had previously identified as basic to the topology of Minkowski spacetime, Savitt (2009) called it the Alexandrov present, or ALEX (e_1, e_2) for short, where e_1 and e_2 are two events on timelike curve or worldline λ, with e_1 earlier than e_2: "I propose that ALEX (e_1, e_2) is the present for the interval from e_1 to e_2 along λ." (Savitt 2009, 357).[51]

In his (2015) Savitt prefers to call the Alexandrov present a *causal diamond*, since this is the term used in the physics literature for the Alexandrov interval. As

[51] Steve Savitt and I discovered we were working along similar lines in an email exchange in 2003. My draft of the paper I was working on then became (Arthur 2006); Steve's draft paper "Time Travel and Becoming" (my copy is dated November 26, 2004) was eventually incorporated into later published papers, like his (Savitt 2009).

he notes, some physicists define a causal diamond as including the boundary of the Alexandrov interval, that is, as the closed set of points in the future of some point p and in the past of another q, while others prefer to exclude the boundary. The above definitions would include processes on the boundary, since a continuous, future-directed lightlike curve from p to q would represent the paths of a process (such as a light ray) going from p to q at the limiting velocity c, and such a process would be absolutely after p and before q. Since such a process could not interact with the process whose worldline extends from p to q, however, it is preferable not to include the boundary in the definition of the Alexandrov present.

Accordingly, in his (2015) Savitt defines his causal diamonds as excluding the boundary. For a spacetime that is "strongly causal" (we'll return to that in Chap. 7) and temporally oriented, then if $I^+(p)$ is the set of points reachable from p by a continuous, everywhere future-directed timelike curve, and $I^-(q)$ is the set of all points from which a continuous, everywhere future-directed timelike curve can reach q, then "the set $I^+(p) \cap I^-(q)$ is a causal diamond" (Savitt 2015, 19, n.3). Savitt discusses the use of these diamonds by theoretical physicists in a variety of contexts, and notes that some cosmologists use diamonds for which the proper time separation between p and q is not small compared to the curvature of spacetime.

Savitt's conception of the present in relativity, however, is not quite the same as mine, and this may have been the source of some confusion in the literature. One thing I omitted above from his definition of the Alexandrov present in his (2009) is that his two events e_1 and e_2 "are 1 s apart on a timelike curve, λ" (2009, 356). Savitt chooses this interval of 1 s for convenience, as "a middling value" between the upper (0.5 s) and lower (3 s) bounds of the specious present suggested by "an eminent psychologist" (359). That would make the present relative to the specious present of human observers. On my construal, on the other hand, the present is indexed to a small section of a worldline that is "usually" of a short proper duration and that *could be* that of "an observer considering", but need not be.[52] For, as I wrote of this construal of the present,

> although it can accommodate what may be present to an observer's conscious experience—and thus preserve our intuitions about the great extent of the present at any moment of consciousness—it does not depend on it. *Any segment of a worldline will have a region of spacetime that is present to it* according to this definition. (Arthur 2006, 148)

According to this construal it makes sense to talk of today or this year as "the present", and to refer to the "present epoch" even in a cosmological context. In each case the present will comprise all those extended events or processes with which the Earth may be said to be in interaction within the proper duration concerned. This is why I called it the "interactive present" in my (2006). Within such a large present all kinds of events can have become and can be in the process of becoming. Savitt, however, "thinks of the present as a locus of becoming, rather than as the 'cutting edge' of what has become" (2015, 23), so that passage is conceived by him as consisting in a

[52] Thus Savitt's claim that "Arthur (2006) and I (Savitt, 2009) proposed that the present in (time-oriented) Minkowski spacetime should be thought of as a small causal diamond" (Savitt 2015, 19) is not quite accurate: for me it is not necessarily small.

succession of presents, each one a localized region of spacetime in which becoming occurs. I'll return to this below.

First, however, we should be clear about what purpose the Alexandrov present is intended to serve. Stein's account of presentness in terms of interaction is intended to explain why the present appears to be short in duration and yet large in spatial extent. This depends on the finiteness and the large value of c (in terms of human centred units such as the metre and the second), but not on any features specific to relativity theory. It explains what Whitehead called "presentational immediacy" (Whitehead 1930, 13), if that is understood as the feeling of the immediacy of certain processes experienced as present to us, without assuming (with him) that all processes we regard as present are present to each other in some objective moment. The idea of such an absolute simultaneity is an inferred concept in classical physics, as we saw in Sect. 6.2, and goes beyond what is actually experienced. The virtue of Stein's account is to show how it is not necessary to presume it in order to account for the origins of our intuition of "presentational immediacy". Savitt and I had followed Stein in arguing that the endowing of the present of human awareness with a short duration allows us to account for intuitions of the present being of large spatial extent, and had proposed that an extension of this into relativistic physics required this extent to be the spatiotemporal region given by the Alexandrov interval or causal diamond.

Here it must be stressed that the Alexandrov present is not intended to reproduce all the features of the present of untutored common sense, the phenomenal present. As Mauro Dorato has observed, not only would such a phenomenal present include massively distant events (such as the supernova observed as happening by Chinese astronomers in 1054 CE, mentioned at the beginning of this chapter), it would not include events to which we were not paying any attention, or point-events which were merely possible and not actual.[53] Nevertheless the interactive present is adequate to explain why we perceive certain events happening at a distance (even thousands of kilometres away) as being present, and explains our intuition of the "reality" or robustness of present events in the sense of their being "at hand".[54]

Third, the Alexandrov present is not proposed in order to explain becoming. It presupposes—as is implicit in Robb's work and made explicit by Stein—that becoming takes place along worldlines, and not by the advance of a world-wide now or plane of simultaneity. As Savitt notes, I myself had argued this in an earlier paper (Arthur 1982),[55] and such a conception is endorsed by Dieks, Dorato, Myrvold and

[53]Most of these points were made in my article: "The interactive present is not the same, however as the *passive* or *subjective present*, the set of all those events of which we are consciously aware at the moment of considering them" (Arthur 2006, 151). Dorato acknowledges this in his (2011), so his criticism of our proposal on these grounds seems to involve some misunderstanding. Cf. Savitt's response: "Dorato is surely right when he says that causal diamonds, if proposed as a scientific successor concept to our common sense concept of the present, do 'not correctly pick out the events we intend to pick out when we use 'now' in ordinary language.' It is true, however, that nothing in *M* does." (Savitt 2015, 21).

[54]See Savitt's (2015) for a detailed responses to this and other objections raised by Dorato in his (2011) to the "Arthur-Savitt" conception of the present.

[55]"Suppose we then conjecture the following (Arthur 1982: 107): 'It is this proper time which is understood to measure *the* rate of becoming for the possible process following this timelike line

others.[56] Thus, on the view I am advocating, events come to be in the present in a quite specific sense: if one temporally extended event *b* lies in the Alexandrov present of another *a*, then *b* comes about during the proper time of the event *a*. Such a notion is neither symmetric (even though a part of *a* will lie in *b*'s present, and vice versa) nor transitive. So there is no question of this construal supporting the notion of the present as an equivalence class of events separating the past from the future.

This is relevant to an objection raised by Craig Callender to Stein's conception of the relation of "having become for". He claims that "the thinnest requirement one might put on becoming" is the condition that "at least one event in the universe shares its present with another event's present" (Callender 2000, S592). As we have just seen, this requirement is not satisfied by the above account, where one extended event may share a great deal of its Alexandrov present with another, but cannot share exactly the same present. But Callender is thinking of point-events, and he offers the above characterization as a gloss on a condition that he insists must be satisfied by any relation R*xy*, "*x* has become for *y*", where *x* and *y* are point-events in Minkowski spacetime. This is what he calls the "non-uniqueness condition", NU, defined as $\exists x \exists y \exists R(Rxy \ \& \ Ryx \ \& \ x \neq y)$. Now, Stein's relation B*xy* is provably antisymmetric: that is, for any point-events *x* and *y*, if *x* has become for *y* and *y* has become for *x*, then *x* and *y* are identical. NU simply asserts the existence of events that contradict this property. But why should we accept NU? And why should we believe that NU encodes the property of sharing a present that Callender believes is essential to R?

It would seem that this derives from Stein, who (hypothetically) asserts the equivalence of "sharing a present" with NU in the antecedent of a conditional he states in his article of 1968: "if 'presentness to each other' of events is taken to mean that *each has for the other already become*, then … in Einstein-Minkowski space-time *an event's present is constituted by itself alone*." (Stein 1968, 15). This follows from the anti-symmetry property of his relation B*xy*. But as I understand his argument, Stein is not arguing for the truth of antecedent of the stated conditional statement, but rather that its consequent—namely Robb's construal of the present—is untenable. And granting the negation of its consequent, the conditional entails that compresence (the presentness of events to each other) should *not* be understood to mean that each has for the other already become. Hence the need to adopt a different conception of compresence in terms of the possibility of mutual interaction between temporally extended events. The extended events contained in ALEX (e_1, e_2) will have become for e_2, but not for e_1. They will be becoming while (e_1, e_2) is in process, but will not wholly have become until e_2.

Now if one insists on linking becoming with being in the present, then one could regard the whole region of ALEX (e_1, e_2) as the locus of becoming for the extended event (e_1, e_2). Becoming would consist in a succession of presents or causal diamonds *D* aligned along a worldline. If this is what Savitt means by his characterization of the

(or *worldline*)'. It is this idea of becoming along a timelike line, local becoming, that underlies my negative evaluation of the two anti-passage arguments presented above." (Savitt 2011, 11).

[56] See Dieks (2006), Dorato (2006a), Myrvold (2003, Sect. 2).

present as "a locus of becoming" (2015, 23), then his indexing of *D* to a proper time of about 1 s acquires an ontological significance lacking in my characterization of the present as simply relative to a segment of a worldline. For it implies that becoming takes place in a process in a small region of spacetime corresponding to 1 s of its proper time, but that events cannot be said to have become until after that proper time has elapsed. Events are in the process of becoming in each *D*, but nothing has become until after each *D*.[57] This is somewhat evocative of Bergson and Whitehead, for whom the continuous, extensive, linear time of mathematical physics is an after-effect of real becoming (*la durée réelle*), which happens in "indivisible, temporal pulsations".[58] But Bergson's *la durée* is "immediately perceived duration", and we have said enough about the mistake of attempting to base physical time on subjective experience, which is no part of my or Savitt's remit.

Turning to physical time, as opposed to the time of psychology, the interval of 1 s appears arbitrary, as Dorato has objected (2011, 302). But the examples Savitt gives of the use of causal diamonds in contemporary physics suggest that a finite time-indexed present might have application at much smaller time scales, where quantum considerations would come into play. There a discrete temporal extent is motivated, not by considerations of its relationship to human perceptions, but by considerations of the indeterminateness of becoming below a certain threshold. But before we enter into quantum considerations, we have first the small task of digesting Einstein's second revolution in our concepts of space and time, wrought by his theory of General Relativity!

As I shall try to show, however, there are problems in the interpretation of both general relativity and quantum theory that arise from an inadequate appreciation of the lessons of special relativity. As Savitt has said, the most important and surprising of such lessons is that temporal passage "is a *local* phenomenon, tied to a world line", and not tied "to an advancing *global* now". It is the latter idea, "buried deep in our worldview, … that we must transcend" (Savitt 2011, 14).

6.6 Summary

- This chapter treated the topic of the status of the present in the face of the relativity of simultaneity of SR. Beginning with a discussion of the status of the present in classical theory, I argued that the events appearing in phenomenal experience are all events that occurred in the past. This already creates difficulties for the kind of primacy accorded to the events of present experience, as in Whitehead's

[57] Savitt provides a definition of becoming relative to a causal diamond *D* in his paper (2015, 23).

[58] Here I am quoting Milič Čapek's synthesis of the views of Bergson and James: "Mathematical continuity is thus nothing but *discontinuity endlessly repeated*, which ignores the natural articulation of duration into indivisible, temporal pulsations" (Čapek 1971, 139). This is similar to Whitehead's atomic becoming, discussed in Chap. 2. Bergson himself characterizes *la durée* as "uninterrupted transition, multiplicity without divisibility and succession without separation" (Bergson 1965, 44–45).

idea of "presentational immediacy", and necessitates a distinction between the phenomenal present and the inferred present. An easily visualizable illustration of the distinction is given using the observations of time that would be made by the twins of SR's Twin Paradox according to classical theory.

- One of the main responses to the difficulty of the relativity of simultaneity in SR has been to cast the present as *observer-dependent*. Insofar as this involves conceiving the universe as a static block and making becoming wholly subjective, this has already been treated in Chap. 3. But a second contributing factor to this view has been the idea that observers have their own reference frames, and that time lapses differently for them because they see different events as present. In order to untangle the misconceptions here, I continued the illustration of the twin paradox to show what the twins would perceive, what they would infer, and how this is all mutually consistent, without assuming any observer-dependence of becoming. This reinforces the conclusion of Chap. 5 that time lapse is perfectly objective in SR, and measured by the invariant proper time, and not by the frame-relative time coordinate function.
- Consequently the temporal ordering of point-events in Minkowski spacetime is not a total ordering, but a partial ordering, which A. A. Robb in 1911 termed a "conical order": not every two point-events occur in an order of succession. Rather all and only those events that are connectible by a physical influence are earlier or later—Robb's postulate of Chronological Precedence (CP). From this Robb inferred that there was no temporal relationship between events in space-like separation, and proposed that an event's present does not extend beyond the point of spacetime where it occurs. It was also explained how, from this postulate of CP together with the principle of relativity and certain natural constraints on physical process, Robb was able to construct Minkowski spacetime; a result that was fortified by the contributions of later authors. There is a perfectly well defined spatiotemporal interval between two events that are space-like separated, but no time interval. It was argued that this perspective on SR precludes regarding tachyons as processes.
- Robb's punctual present is hard to square with the robustness of our sense of the present as having some extension. I suggest a way of defining a notion of the present that is compatible with special relativity, and also with the temporal extendedness of events, such as that of experiencing an event. Such a present is relative to segment of a worldline, and is comprised by a four dimensional region of Minkowski spacetime contained within the two hypercones centred on that segment. I argued that such a notion could account for the compresence of certain enduring things, although it would not be identical with either the phenomenal present or the inferred present of classical physics.

Appendix: Informal Sketch of a Group-Theoretic Derivation of the Lorentz Transformations[59]

The basic idea is this. Inertial frames are characterized not by rods and clocks, but by Lange's construction (Lange 1886), where bodies moving relatively to one another in straight lines without circular motion produce equal displacements in equal times. Such motions form an equivalence class, and this class defines an inertial frame. The Principle of Inertia is then constituted by the assertion that there exists at least one such inertial frame. The Principle of Relativity, in its general form, then asserts the equivalence of these inertial frames for expressing the laws of physics; more precisely, that there exists an infinite class of equivalent inertial frames in relative motion one to another, such that the laws of mechanics are invariant under transformations from one inertial frame to any other. The idea is to derive the various possible forms of this group of transformations by imposing certain plausible constraints on them. A succinct way of achieving this that brings out some essential features of Robb's construction may be summarized as follows.

The same vector displacement that is produced by a point moving with a uniform velocity in a given inertial frame (which we may term, conventionally, the "stationary frame") may equivalently be produced by a point at rest in a frame moving with the same velocity relative to the original inertial frame, that is, by an active boost in velocity of that frame. We may call this a "boost displacement". The Principle of Relativity can now be understood in group theoretical terms as asserting that these transformations by a boost in uniform velocity v, constitute a group of transformations, more precisely, a differentiable and connected one-parameter group. Now, according to a standard mathematical theorem of group theory, for any such group there exists an *additive parameter,* such as the *angle* for a group of *rotations* in Euclidean space. This parameter must be a function of the velocity, and also have the dimensions of velocity. Classically, it was assumed that it simply *is* the velocity. But given the fact that lengths and times in moving frames are not invariant (due to length contraction and time dilation), we cannot suppose that this holds universally. So as not to prejudge things, following Robb we will call this additive parameter producing displacements by velocity boosts the "rapidity", denoted φ.[60] The additive nature of the parameter means that the rapidity addition law is just like the velocity addition law in classical mechanics, $\underline{\varphi}_{12} = \underline{\varphi}_1 + \underline{\varphi}_2$. It is the analogue of the Parallelogram Law, $\underline{\mathbf{v}}_{12} = \underline{\mathbf{v}}_1 + \underline{\mathbf{v}}_2$, expressed by Newton as follows:

[59] This sketch is based on (Arthur 2007), which in turn was largely inspired by Provost (1980). See these papers for further details.

[60] "We propose now to define what we shall call the rapidity of a particle with respect to a system of permanent configuration … we shall see later that, for large values, it is the rapidity and not the velocity which follows the additive law." (Robb 1911, 8–9).

> A body acted on by [two] forces acting jointly describes the parallelogram in the same time in which it would describe the sides if the forces were acting separately. (Newton 1999, 417)[61]

Here Newton is assuming that each "force" will produce a corresponding inertial motion proportional to its velocity, and that each inertial motion will produce a displacement in a given time proportional to that time.

Now by the Principle of Relativity any displacement $\underline{\mathbf{x}}$ in any one inertial frame can be realized by an active boost of a point-particle at rest in a second inertial frame in motion relative to the first with a rapidity $\underline{\varphi}$. What plausible constraints should there be on these boosts beyond the rapidity addition law? One of these is linearity, which can be shown to follow from the *homogeneity* of spacetime. Second, the boost displacement should obey *space-reflection*, so that that the relative rapidity $\underline{\varphi}$ of two inertial frames changes sign under change of sign of all the spatial axes: if $\underline{\mathbf{x}} \to -\underline{\mathbf{x}}$, $\underline{\varphi}(-\underline{\mathbf{x}}) \to -\underline{\varphi}(\underline{\mathbf{x}})$. Third, they should obey the same *equal times principle* as that holding for displacements in the stationary frame according in Newton's Parallelogram Law: if three boost displacements $\underline{\mathbf{x}}_1, \underline{\mathbf{x}}_2, \underline{\mathbf{x}}_3$ satisfy $\underline{\mathbf{x}}_1 + \underline{\mathbf{x}}_2 = \underline{\mathbf{x}}_3$, then the displacements on both sides of the equation have been realized in equal times. Fourth, we assume *causality*, which here consists in the requirement that there are (at least some) time intervals which do not change sign under any boost transformation. This last constraint rules out a group of transformations which are such that one may always find a boost that changes the sign of the time interval. In Galilean spacetime, all time intervals satisfy this constraint; in Minkowski spacetime, it is satisfied by pairs of events separated by a timelike or lightlike interval. So in those cases the constraint is therefore equivalent to Robb's assumption that there are asymmetric relations of chronological precedence that are absolute (independent of frame).

Expressing these constraints in group theoretical terms, one can prove that there are only two possibilities consistent with all of them (which we can express in one dimension without loss of generality). The first just returns $\varphi = v$, rapidity is identical with velocity, as was assumed classically. From this we obtain the Galilean transformations consistent with the classical velocity addition law, $\underline{\mathbf{v}}_{12} = \underline{\mathbf{v}}_1 + \underline{\mathbf{v}}_2$. The second and more interesting case delivers $\varphi = k \tanh^{-1}(v/k)$, or equivalently, $v = k \tanh(\varphi/k)$, with rapidity adding according to $\varphi_{12} = \varphi_1 + \varphi_2$. Here k is an undetermined constant with the dimensions of velocity, whose value is determined experimentally to be equal to c, the speed of light in a vacuum. Notably, this does not need to depend on anything to do with light. It would suffice, for instance, to determine that at a relative velocity of 3/5 of the value of c there is a time dilation of some process by a factor of 5/4. Taking units where $c = 1$, as is customary, rapidity is given by $v = \tanh \varphi$, in agreement with Robb's definition.[62] As he explained there, this gives the Lorentz transformations in the following succinct form:

[61] A more literal translation would be: "A body [carried] by conjoined forces describes the diagonal of a parallelogram in the same time as [it would] the sides by the separate forces." Newton assumes that the forces and velocities are directed, so that his law is a law of vector addition, *avant la letter*. For simplicity's sake, we may consider it in one dimension only without loss of generality.

[62] "If v be the absolute velocity of the particle with respect to the system, then the inverse hyperbolic tangent of v will be spoken of as the rapidity. Thus if ω be the rapidity, $v = \tanh \omega$. As ω increases

$$x' = x\cosh\varphi - t\sinh\varphi \quad \text{—equivalent to } x' = \gamma(x - vt),\ \gamma = 1/\sqrt{(1 - v^2)}$$
$$t' = t\cosh\varphi - x\sinh\varphi \quad \text{—equivalent to } t' = \gamma(t - xv),\ \gamma = 1/\sqrt{(1 - v^2)}$$

What we have, then, is a bifurcation of the classical velocity concept into two distinct concepts: the *velocity* in a given inertial frame of reference, and the *rapidity* characterizing active transformations consisting in boosts in instantaneous velocity (or, equivalently, passive transformations from one inertial frame to another in relative motion to it).[63] We may say that the classical concept of velocity is degenerate, and that this degeneracy is only revealed as velocities become an appreciable fraction of c. This is analogous to the degeneracy of classical time noted in the previous chapter, where the classical time concept that did double duty in keeping track of synchrony and tracking becoming, at velocities close to c bifurcates into co-ordinate time (keeping track of synchrony) and proper time (tracking becoming).

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from 0 to ∞, v increases from 0 to 1. For small values of ω we have, practically, velocity is equal to rapidity, but we shall see later that, for large values, it is the rapidity and not the velocity which follows the additive law." (Robb 1911, 8–9).

[63] As Lévy-Leblond and Provost (1979, 1045) remark, it shows the need "to replace the Galilean velocity by two separate concepts: 'velocity' *v,* as expressing the time rate of change of position, and 'rapidity' φ, as the natural additive group parameter."

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